

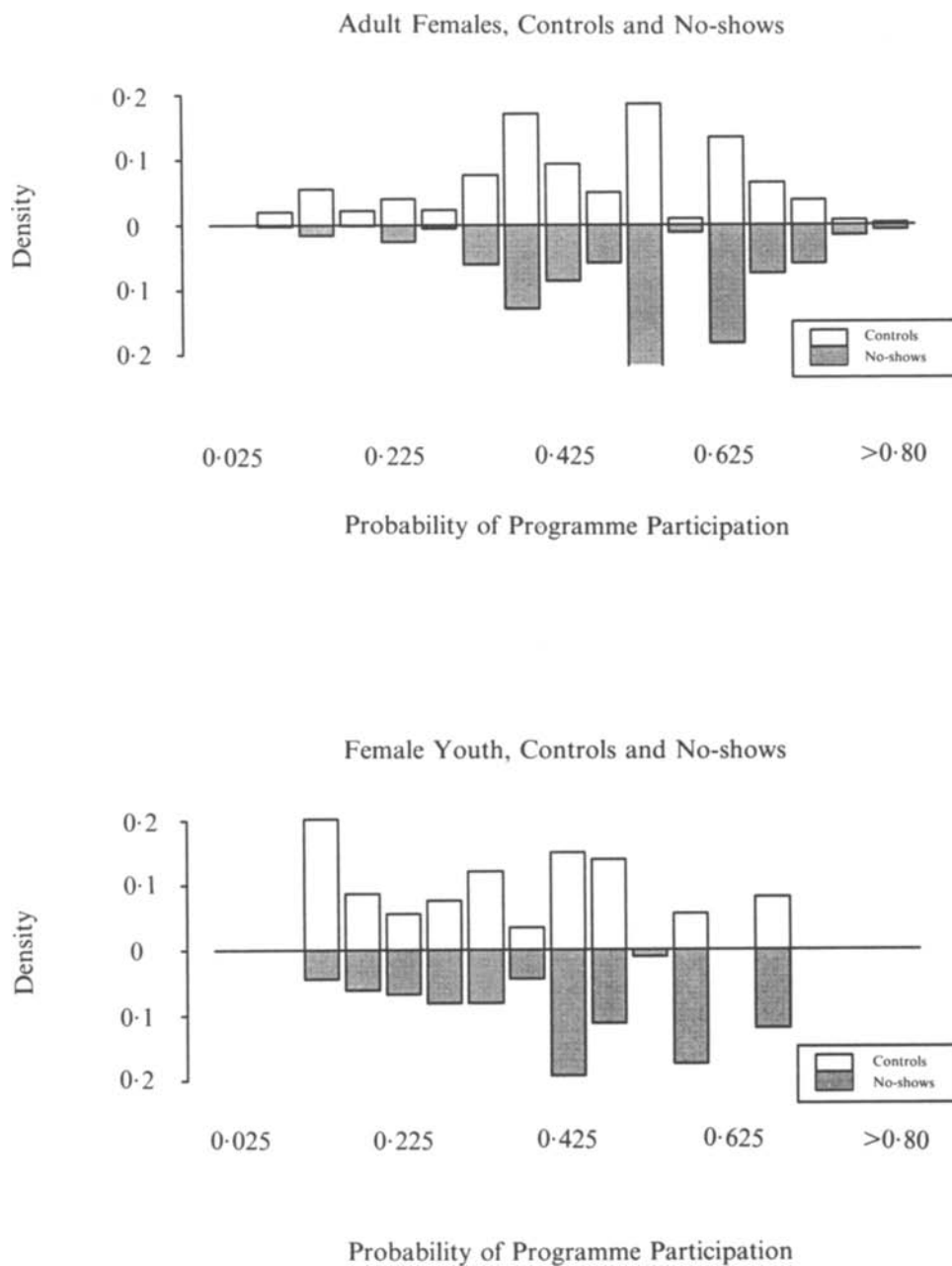


FIGURE 2. continued.

Little of a general nature can be concluded from our evidence about the use of no-shows. As programmes are improved, the proportion of no-shows would be expected to decline. No-shows would then become a more select sample. Nonetheless, the lower biases estimated for adult no-show comparison groups confirm the conclusions reached in the analysis of ENP-control samples. A major component of what is commonly regarded as evaluation bias is due to mismatch on observables. Picking comparison groups of persons in the same labour market, administering them the same questionnaire, and weighting the comparisons comparably goes a long way toward replicating the estimates produced from a social experiment.

16. CONCLUSION AND DISCUSSION OF RELATED WORK

This paper examines various matching methods and extensions of matching methods for evaluating job training programmes using several different sources of nonexperimental data combined with experimental data. Our approach to evaluation has two main components: (a) estimating a model that successfully predicts who participates in the programme; and (b) using the estimated probability of participation in matching and extensions of the matching method. The estimators that we propose in this paper and rigorously justify in Heckman, Ichimura Todd (1997) perform well among the estimators we examine here. Their application to the evaluation of a major job training programme produces impact estimates and inferences "fairly close" to those produced from a randomized evaluation of the programme, although estimated selection bias is still a substantial fraction of the experimentally-estimated programme impacts.

We determine that a regression-adjusted semiparametric conditional difference-in-differences matching estimator often performs the best among the class of estimators we examine, especially when omitted time-invariant characteristics are an important source of bias. Placing nonparticipants in the same labour market as participants, administering both the same questionnaire, and weighting their observed characteristics in the same way as that of participants, produces estimates of programme impacts that are fairly close to those produced from an experimental evaluation. We find that for adult males, selection bias due to differences in unobservables between participants and non-participants is a relatively small component of total evaluation bias as conventionally measured (B). For other groups, selection bias is a larger component of bias as conventionally measured. If a few simple empirical principles are followed, one can dramatically improve on the poor performance of nonexperimental estimators emphasized in LaLonde's (1986) influential paper. Our evidence also indicates that a substantial fraction of the bias reported by LaLonde was due to questionnaire mismatch and geographic mismatch and not self-selection bias generated from a common distribution of unobservables across comparison and treatment groups. It is the distributions of the characteristics that appear to be different across groups.

A major issue that arises in the application of matching in ordinary observational analyses is determining if the assumptions that justify matching estimators are satisfied. In Heckman, Ichimura and Todd (1996b), we confront this issue and propose a test of the validity of matching under the assumption that there is an exclusion restriction. Letting $X = (T, Z)$, we assume that there is some matching variable in Z not in T . Then, if Assumption (A-4') is true, $E(Y_1 | T, P(Z), D = 1) - E(Y_0 | T, P(Z), D = 0)$ should not be a function of the Z variables excluded from T provided that Z does not predict from the programme for participants. Using kernel estimates of the two conditional expectations, we develop a formal test of this hypothesis and related hypotheses.

Matching and our extensions of it differ from the widely-used instrumental variables estimator. In our context, the conventional instrumental variables estimator is based on the assumption that $E(U_0|X, P, D=1)=0$ and $E(U_0|X, P, D=0)=0$.³⁶ Elsewhere we have shown that using our estimated P as an ordinary instrument produces substantially biased estimators of programme impact. (Heckman (1995).) Our evidence does not justify application of the instrumental variables estimator to our data. The matching estimator assumes that P balances the bias, $E(U_0|P, D=1)=E(U_0|P, D=0)$, but does not assume that either term equals zero. Similarly, the conditional difference-in-differences estimator introduced in this paper also assumes bias is balanced in the differences, either in levels or residuals, but does not equate conditional means to zero.

A major finding of this paper is that comparing the incomparable—i.e. violating the common support condition for the matching variables—is a major source of evaluation bias as conventionally measured. A major limitation of the non-experimental method is that the support in comparison samples may be very different from the support in control samples. Restricting the application of nonexperimental methods to regions of common support may change the parameter being estimated in a non-experimental evaluation from what is estimated by an experiment. ($M(S_{10}) \neq M(S_E)$ where S_{10} is the common support in a nonexperimental evaluation and S_E is the support in the benchmark experimental evaluation). The evidence reported in Table 4 demonstrates that this source of bias is substantial for certain demographic groups. In our data, restricting S to the common subset to balance the bias produces estimates of mean programme impacts that are uniformly higher than the experimentally-determined estimator for the full support of programme participants.

As is true of any empirical study, our findings may not generalize beyond our data. However, it is important to note that the proposed two stage evaluation strategy performs comparably in its application to four distinct demographic groups. Moreover, the main features of the JTPA programme are common to many other job training programmes both in the U.S. and elsewhere. Participants in the JTPA programme have the same pre-programme dip in earnings as found in numerous job training programmes and it is likely that similar factors operate in generating selection decisions in all of these programmes. The range of services offered is comparable to other programmes. (Heckman (1995).) Thus it is likely that the insights gained from our study of the JTPA programme on the effectiveness of different estimators also apply in evaluating other training programmes targeted toward disadvantaged workers.

APPENDIX

A. Test statistics and variance estimators

In this appendix, we discuss the details of the tests for conditional independence, conditional mean independence, given by hypotheses $H_{(A-3)}$, $H_{(A-3')}$, $H_{(A-4)}$, and $H_{(A-4')}$ in the text as well as the hypotheses concerning the validity of the conditional differences-in-differences method, $H_{(A-5)}$ and $H_{(A-5')}$. We first give test statistics for use in a single cross-section and then present joint tests across multiple time periods, allowing for the possibility of unbalanced panel data.³⁷

Testing for conditional independence ($H_{(A-3)}$ and $H_{(A-3')}$):

36. Heckman and Smith (1997) compare matching and IV methods.

37. The test statistics for conditional mean independence were proposed in Heckman, Ichimura, Smith and Todd (1996b).

To test for $H_{(A-3)}$ at a point $Y_0 = y_0$ conditional on $P = p$, we estimate the conditional c.d.f. $\hat{F}_d(p) = \hat{F}_d(y_0 | P = p, D = d)$, for $D = 0$ and $D = 1$ groups using local linear regression smoothing methods.³⁸ The estimator $\hat{F}_d(p)$ asymptotically satisfies

$$(\hat{F}_d(p) - F_d(p)) \sim N(B_d, V_d),$$

where

$$B_d = \frac{1}{2} F_d''(p) \frac{C_2}{C_1} a_{N_d}^2 \quad \text{and} \quad V_d = (N_d a_{N_d})^{-1} \frac{\text{Var}(\varepsilon_{id} | D = d, P = p) C_3}{f_d(p) C_1}.$$

B_d is a bias term. $F_d''(p)$ is the second derivative of the conditional c.d.f. with respect to p evaluated at p , $f_d(p)$ is the density of P evaluated at p , $\varepsilon_{id} = 1(Y_{0i} \leq y_0) - F_d(P_i)$, N_d is the number of observations, and a_{N_d} is a bandwidth that converges to zero as N_d gets large and satisfies $N_d a_{N_d} \rightarrow \infty$, $N_d a_{N_d}^5 < \infty$.³⁹ The terms C_1 , C_2 , and C_3 are constants that depend on the kernel function $G(\cdot)$ used in the local linear regression:

$$\begin{aligned} C_1 &= \int s^2 G(s) ds \int G(s) ds - \left[\int s G(s) ds \right]^2, \\ C_2 &= \left[\int s^2 G(s) ds \right]^2 - \int s^3 G(s) ds \int s G(s) ds, \\ C_3 &= \int \left[\int s^2 G(s) ds - u \int s G(s) ds \right]^2 G^2(u) du, \end{aligned}$$

where the integrals are evaluated over all $s \in (-\infty, \infty)$.⁴⁰ The kernel function used to obtain all estimates reported in this paper is the biweight kernel

$$\begin{aligned} G(s) &= 15/16(s^2 - 1)^2 \quad |s| < 1; \\ &= 0 \quad |s| \geq 1. \end{aligned}$$

For testing $H_{(A-3)}$, the test statistic is

$$(\hat{F}_1(p) - \hat{F}_0(p))'(\hat{V}_0 + \hat{V}_1)^{-1}(\hat{F}_1(p) - \hat{F}_0(p)) \sim \chi_1^2.$$

Under the null, if the same bandwidth is used to estimate $\hat{F}_1(p)$ and $\hat{F}_0(p)$ then the bias expressions cancel so only estimators for the variance are needed.⁴¹ Consistent estimators $\hat{V}_1$ and $\hat{V}_2$ of V_1 and V_2 are presented below. Test of conditional independence of residuals ($H_{(A-3)}$) are analogous, except that they replace Y_{0i} by $Y_{0i} - X_i \beta = \hat{U}_i$.

Testing the conditional mean independence assumption ($H_{(A-4)}$, $H_{(A-4')}$)

The test of hypothesis $H_{(A-4)}$ is similar to that of $H_{(A-3)}$. First estimate $\hat{m}_1(p) = \hat{E}(Y_0 | P = p, D = 1)$ and $\hat{m}_0(p) = \hat{E}(Y_0 | P = p, D = 0)$ by a local linear regression of Y_{0i} on P_i evaluated at p . $\hat{m}_d(p)$ is distributed as

$$(\hat{m}_d(p) - m_d(p)) \sim N(\tilde{B}_d, \tilde{V}_d),$$

where

$$\tilde{B}_d = \frac{1}{2} m_d''(p) \frac{C_2}{C_1} a_{N_d}^2 \quad \text{and} \quad \tilde{V}_d = (N_d a_{N_d})^{-1} \frac{\text{Var}(\varepsilon_{id} | D = d, P = p) C_3}{f_d(p) C_1}.$$

38. For a given y_0 we perform a nonparametric regression of the indicator variable $1(Y_i \leq y_0)$ on P_i for $D = 0$ and $D = 1$ groups and evaluate the function at p . A fixed bandwidth equal to 0.06 is used in the test.

39. See Heckman, Ichimura, Smith and Todd (1996c), Theorem 5, for why these restrictions are needed on the bandwidth.

40. Here we assume p is an interior point of the support.

41. If kernel regression is used instead of local linear regression, the bias terms would not cancel and the test statistic would have a noncentrality parameter. Thus, using local linear regression, with a common bandwidth for estimating each of the nonparametric functions, simplifies the test because it yields a test statistic that is centrally distributed.